Thermodynamic Systems

A system can be defined as a collection of matter with identifiable boundaries.

A thermodynamic system can be defined similarly, with identifiable boundaries that are flexible (can expand or contract), with work and heat being able to move across the boundary; to and from the surroundings to the system.

Hence, the thermodynamic system definition is a system in which energy in various forms is transferred or converted.

When analysing and defining systems, it is important to define what is the system, and what is the surroundings, where the system is the section that we want to study, and the surroundings is everything else that may influence the system. A system may not necessarily have solid bounds and the system boundaries may be defined differently.

For this module, the fluid itself is usually the system, while the solid walls, pistons, and other components are usually the surroundings.

Open & Closed Systems

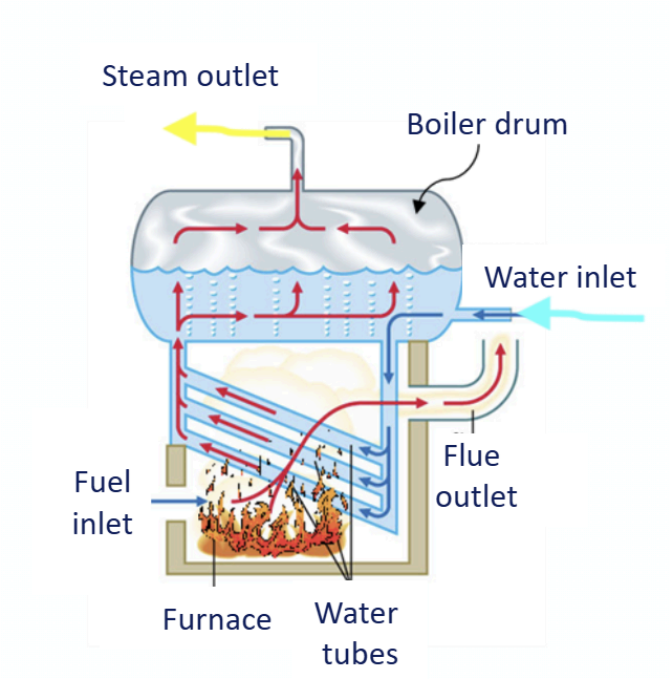
- Systems can be either open or closed, referring to whether mass can flow in and out of the system.
- Closed systems refer to systems that do not allow mass to flow in and out, however energy can pass through and volume of the system can change. Processes in these systems are known as non-flow processes.
- Open systems refer to systems that do allow mass to flow in and out, allowing mass and energy to pass through the system. Hence, processes in these systems are known as flow processes. Most practical systems are usually open systems.

Common Thermodynamic Systems

- Heat Engines
  - A heat engine is a commonly used device for converting heat into work, used in power generation, cars, planes.
  - Usually, heat engines are quite complicated, and can be broken down into simpler thermodynamic systems:

Steam Boilers

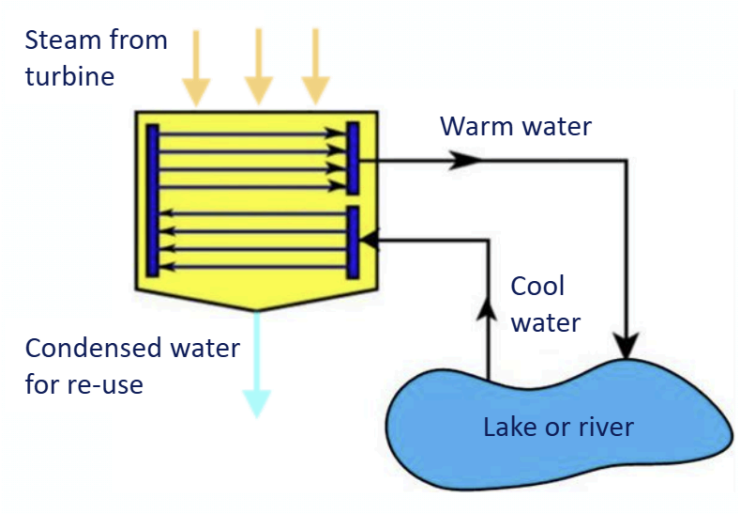
A boiler heats water to make steam by burning fuels for thermal energy.



These boilers have two main parts, the rack of tubes, and the drum. Water flows from the drum into the tubes, where they are heated in the furnace, causing them to boil and turn into steam, which rises and flows into the drum, and out of it to the next component.

Steam Condensers

A condenser cools steam into water. They are similar to boilers, in that they have a rack of tubes, where the cool water passes through to allow the steam to condense on it.



Turbines

A turbine extracts kinetic energy from a moving liquid. The liquid pushes on the blades of the turbine, which causes the turbine and the shaft to rotate.

Nozzles & Diffusers

Nozzles and diffusers are changes in the diameter of tubes.

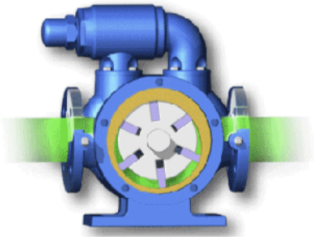
A decrease in diameter is a nozzle, which causes an increase in flow velocity, while an increase in diameter is a diffuser, causing an decrease in flow velocity.

Pumps & Compressors

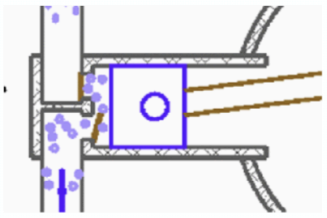
Pumps and compressors transport and pressurise fluids by using external energy (via a motor). These are opposite to turbines, where a turbine outputs work, while pumps/compressors requires work input.

The difference between pumps and compressors is that pumps transfer liquids to different locations, while compressors transfer gases by pressuring a low pressure gas to a high pressure gas, to move it to the next component.

Centrifugal pump



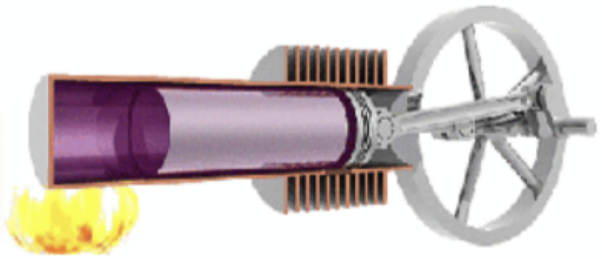
Rotary vane pump



Reciprocating compressor

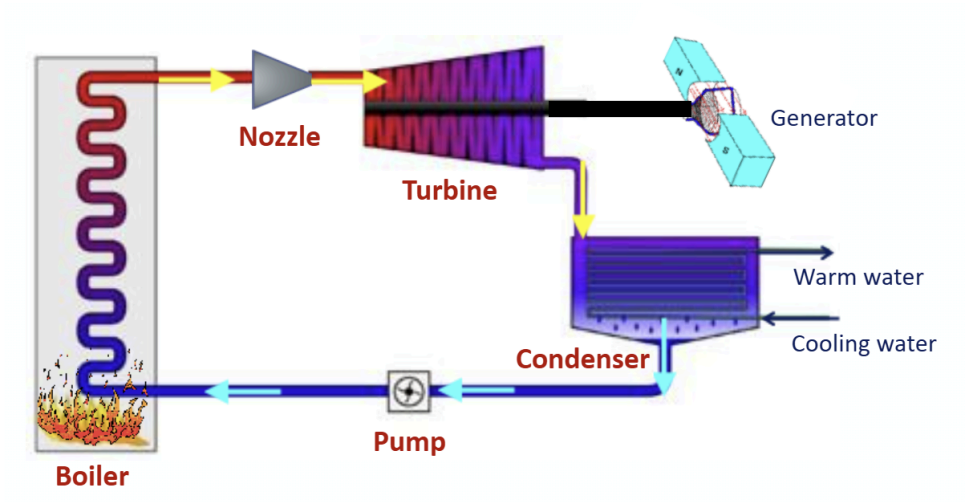
Piston Cylinder Mechanisms

A piston cylinder mechanism comprises of a piston that is able to reciprocate inside a cylinder. The piston is able to move due to pressure of the fluid inside the cylinder. Typically, in engines, the pressure in the fluid is increased by heating the fluid, causing the piston to reciprocate. The reciprocating piston is used to rotate a crankshaft, converting the motion to rotation.



Steam Turbine Engines

Shown is the typical assembly of a steam turbine engine.



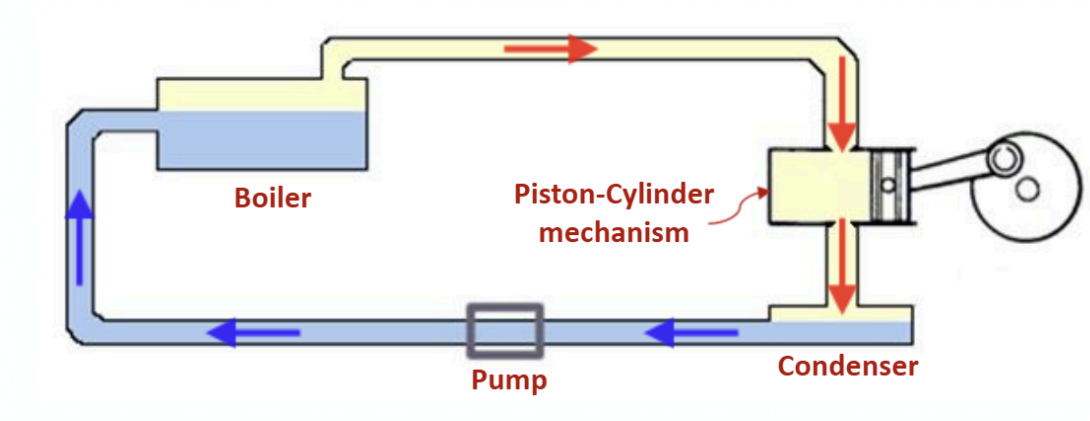
The boiler harnesses energy from the fuel to heat the water into steam, which has the flow velocity increased with the nozzle. Having a higher flow velocity increases the efficiency of the turbine, which turns the turbine, generating energy.

The steam is then sent into a condenser, where it cools down back to water and is pumped to the boiled via a pump to be reused again.

These steam turbine engines are most useful for large scale power generation.

Steam Engines

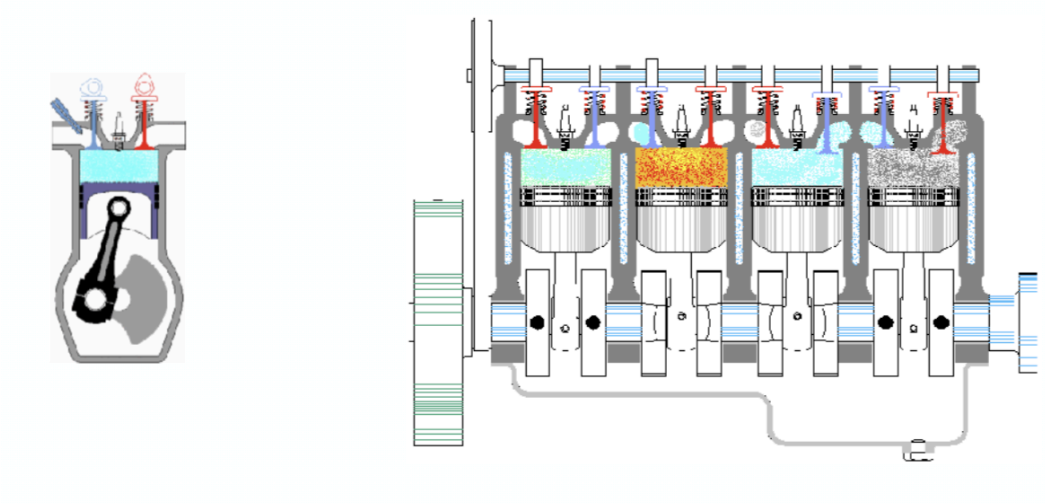
Shown is a typical steam turbine engine



The assembly is very similar to steam turbine engines, however with a piston cylinder mechanism installed instead, as its able to harness energy from the pressurised steam.

Internal Combustion Engines

internal combustion engines make use of multiple piston cylinder mechanisms and gases to output rotational motion.



these engines are typically used in automobiles